

PII Detection and Anonymization Progress Report

COBOL Source Code and Generated Artifacts

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Scope	Person names, IBAN, email, Italian codice fiscale, and COBOL PII-bearing fields
Main engine	Microsoft Presidio
NLP model	spaCy Italian model <code>it_core_news_sm</code>
Custom layer	COBOL-aware recognizers and report generation scripts

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1 Objective

The objective is to identify and anonymize or pseudonymize personal data that may appear inside COBOL source files, copybooks, comments, display messages, string literals, and generated artifacts.

The current PII categories in scope are:

- Person names and surnames
- IBAN values
- Email addresses
- Italian codice fiscale values
- COBOL fields that may carry personal data, such as NOME, COGNOME, MATRICOLA, CODDIP, and IBAN

2 Tools and Components

2.1 Microsoft Presidio

Microsoft Presidio is used as the main PII detection engine. It provides built-in recognizers for structured PII and integrates with NLP models for natural-language entities.

Entity	Purpose
EMAIL_ADDRESS	Detects email addresses.
IBAN_CODE	Detects valid IBAN values.
IT_FISCAL_CODE	Detects Italian codice fiscale values.
PHONE_NUMBER	Detects phone-number-like values.
PERSON	Detects person names through NLP.
LOCATION	Detects locations through NLP.

Presidio is installed locally and can run offline after installation.

2.2 spaCy Italian Model

The installed model is:

```
it_core_news_sm
```

This model is used by Presidio for natural-language entity recognition, especially **PERSON** and **LOCATION**.

2.3 Custom COBOL Recognizers

COBOL source code differs significantly from normal prose. Names and PII may appear in field names, comments, string literals, **DISPLAY** messages, and file assignment values. For this reason, custom COBOL recognizers were added in:

```
scripts/utils/scan_pii_presidio.py
```

Custom Entity	Purpose
COBOL_NAME_LITERAL	Finds COBOL literals assigned to name-like fields.
COBOL_KEY_VALUE_NAME	Finds patterns such as NOME=... and COGNOME=....
COBOL_COMMENT_NAME	Finds likely names in comments, display messages, and string literals. This is an aggressive optional mode.
COBOL_PII_FIELD	Flags fields that may carry PII, such as NOME, COGNOME, MATRICOLA, CODDIP, and IBAN.
IBAN_LIKE	Finds IBAN-shaped values, including synthetic or checksum-invalid test data.
WATCHLIST_PERSON	Finds known names or surnames from a watchlist file.

3 Scripts Created

3.1 PII Scanner

File:

```
scripts/utils/scan_pii_presidio.py
```

Purpose:

- Scans files or folders for PII
- Uses Microsoft Presidio
- Adds COBOL-specific detection rules
- Outputs JSON findings and an HTML visual report

Example command:

```
python scripts/utils/scan_pii_presidio.py artifacts/demo_pii/FAKE_FILE.CBL ^
--out-dir artifacts/demo_pii/fake_file_report ^
--entities PERSON IBAN_CODE IBAN_LIKE EMAIL_ADDRESS IT_FISCAL_CODE ^
COBOL_NAME_LITERAL COBOL_KEY_VALUE_NAME COBOL_COMMENT_NAME
```

3.2 Pseudonymization Script

File:

```
scripts/utils/pseudonymize_pii.py
```

Purpose:

- Replaces detected structured PII with deterministic fake values
- Keeps referential consistency through a secret salt
- Supports IBAN-like pseudonyms and codice fiscale-like pseudonyms
- Supports labeled name fields and COBOL literals moved into name fields

Example command:

```
python scripts/utils/pseudonymize_pii.py artifacts/demo_pii/PDCBVC_fake_pii.CBL ^
--out-dir artifacts/demo_pii/pseudonymized ^
--salt demo-salt
```

4 Test Files

File	Purpose
<code>artifacts/demo_pii/PDCBVC_fake_psi.CBL</code>	Synthetic COBOL file derived from <code>PDCBVC.CBL</code> ; includes fake names, IBANs, codice fiscale values, and email examples.
<code>artifacts/demo_pii/FAKE_FILE.CBL</code>	Synthetic stress test containing many uppercase Italian names in comments, display messages, file names, and literals.
<code>temp/PDHASI06.CBL</code>	Realistic COBOL test file. It contains PII-related fields but no clear embedded IBAN, email, or codice fiscale values.

5 Current Findings

5.1 Structured PII Detection

Structured identifiers are detected well. The most reliable categories are:

- Email
- Valid IBAN
- IBAN-shaped values through `IBAN_LIKE`
- Italian codice fiscale

This is the strongest part of the current solution.

5.2 Person Name Detection

Person-name detection is more difficult. Microsoft Presidio with spaCy can detect normal person names, but COBOL introduces several challenges:

- Names may appear fully uppercase
- Names may appear as single surnames
- Names may appear inside comments
- Names may appear inside `DISPLAY`, `VALUE`, `STRING`, or file assignment literals
- COBOL program names and technical identifiers may look like names

For this reason, custom COBOL-aware detection was added.

5.3 COBOL Comment and Message Detection

The custom entity `COBOL_COMMENT_NAME` was added to detect likely names in:

- COBOL comments
- `DISPLAY` messages
- `VALUE` string literals

- `STRING` statements
- `ASSIGN TO` file names

This mode is intentionally optional because it is more aggressive and may produce false positives in real COBOL files.

5.4 Watchlist Handling

The custom entity `WATCHLIST_PERSON` was added for known names or surnames that must always be detected.

Watchlist file:

```
artifacts/pii_watchlist/names.txt
```

Example entry:

Mattarella

This is useful for known high-risk terms, but it is not a scalable replacement for general name detection.

6 Important Limitations

6.1 Presidio Is Not Perfect for Names

Presidio is strong for structured data but weaker for names in legacy code. It may miss isolated surnames, uppercase names in comments, or names embedded in COBOL messages. It may also incorrectly classify technical identifiers as `PERSON` or `LOCATION`.

6.2 Watchlists Do Not Scale Alone

A watchlist is useful for known names, but it is not realistic to maintain every possible name or surname. The better approach is to combine Presidio, COBOL-aware rules, context-based comment/message detection, optional watchlists, and manual review of HTML reports.

6.3 Automatic Rewriting Requires Care

Detection and pseudonymization should be separated. The recommended process is:

1. Detect PII.
2. Review the HTML report.
3. Decide which entity types should be anonymized.
4. Pseudonymize copied or exported artifacts.
5. Re-scan the pseudonymized output.

Original source files should not be modified directly without review.

7 Recommended Detection Modes

7.1 Strict Values-Only Scan

Use this mode when looking for clear values with fewer false positives:

```
python scripts/utils/scan_pii_presidio.py <input-file-or-folder> ^
--out-dir <report-folder> ^
--entities PERSON IBAN_CODE IBAN_LIKE EMAIL_ADDRESS IT_FISCAL_CODE ^
COBOL_NAME_LITERAL COBOL_KEY_VALUE_NAME
```

7.2 Aggressive Comment and Message Scan

Use this mode when names may appear in comments, display messages, or string literals:

```
python scripts/utils/scan_pii_presidio.py <input-file-or-folder> ^
--out-dir <report-folder> ^
--entities PERSON IBAN_CODE IBAN_LIKE EMAIL_ADDRESS IT_FISCAL_CODE ^
COBOL_NAME_LITERAL COBOL_KEY_VALUE_NAME COBOL_COMMENT_NAME
```

7.3 PII Field Risk Scan

Use this mode when identifying programs that process personal data fields:

```
python scripts/utils/scan_pii_presidio.py <input-file-or-folder> ^
--out-dir <report-folder> ^
--entities COBOL_PII_FIELD
```

8 Next Steps

The next phase is to test different anonymization approaches and compare their accuracy and safety.

Approach	Target	Expected Result
Structured PII pseudonymization	IBAN, email, codice fiscale	High accuracy and low risk of breaking COBOL syntax.
COBOL field-based name pseudonymization	Name-like COBOL fields such as NOME, COGNOME, NOMINATIVO, NOME-REF	Good precision because replacement occurs only in clear field contexts.
Comment and message name anonymization	Names in comments, display messages, and string literals	Better coverage of hidden names, with higher false-positive risk. Requires report review.
Watchlist-based re-placement	Known sensitive names or surnames	Very high precision for known terms, but not sufficient as a standalone solution.

9 Overall Assessment

The current solution is a practical first version for COBOL PII discovery.

Strengths

- Good detection for IBAN, email, and codice fiscale
- COBOL-aware scanning is available
- HTML reports make review easier
- Different scan modes allow strict or aggressive detection
- Runs locally after installation

Weaknesses

- Name detection remains the hardest part
- Aggressive comment scanning can produce false positives
- Automatic anonymization of comments and free text requires review

Recommended position

Use this as a detection and review pipeline first, then apply pseudonymization in controlled modes. The safest production workflow is to pseudonymize copied artifacts, not original source files, and always re-scan the output.